Multi-dimensional Exploration of Evaluating Innovation of Students' Assignment in the Era of Generative Artificial Intelligence

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ABSTRACT

In the era of Generative Artificial Intelligence (GAI), traditional methods of evaluating student assignments face increasing challenges in capturing originality, problem-solving ability, and technical sophistication. This study proposes a structured, AI-driven evaluation framework designed to assess assignment innovation in human–AI collaborative learning environments. The system integrates structured prompt formats, natural language processing (NLP), and multi-dimensional rubrics to measure creativity, logic, problem-solving, and technology application. A controlled experiment involving 120 engineering undergraduates was conducted to compare the performance of students using AI tools versus those working independently. Results show that the AI-assisted group significantly outperformed the control group in multiple dimensions, especially innovation and technical execution. This framework demonstrates scalable, adaptive potential for intelligent evaluation in higher education, while also addressing concerns of fairness, data privacy, and cognitive equity in AI-supported academic work.

Keywords: Generative Artificial Intelligence, assignment evaluation, structured prompt, Human-AI collaboration, creativity assessment, educational NLP, intelligent feedback system

1. INTRODUCTION

1.1 Research Background

The rapid advancement of generative artificial intelligence (GAI) technologies—represented by tools such as ChatGPT, GitHub Copilot, and Claude—has brought transformative changes to education. These tools are increasingly integrated into students' academic routines, reshaping how they complete assignments, engage with knowledge, and demonstrate innovation.

Recent usage data (see Figure 1) shows that over 60% of students regularly use ChatGPT to assist with academic tasks. Among them, 17% report using it consistently, and 24% most of the time. This trend signifies a fundamental shift in the nature of student learning, indicating that human–AI collaboration has become a pervasive and influential component of modern education.

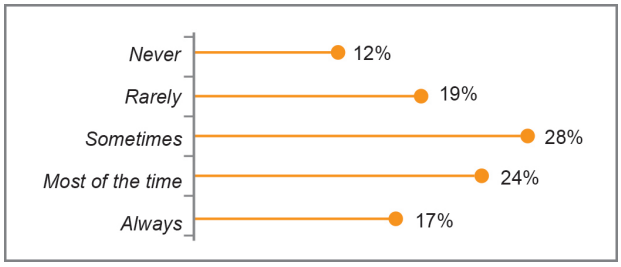


Figure 1. ChatGPT Usage Among Students for Academic Assignments

Source: Intelligent.com. "Survey: How Students Use ChatGPT in School." January 2023.

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The chart illustrates the frequency distribution of ChatGPT usage by university students, highlighting widespread AI integration in coursework.

This evolution poses both opportunities and challenges for educational institutions. On one hand, generative AI can:

- Analyze and evaluate student outputs using natural language processing and deep learning;
- Deliver personalized feedback tailored to each learner's profile;
- Reduce teachers' workloads and increase objectivity in assessment.

On the other hand, these capabilities raise critical concerns, including:

- The erosion of student independence and critical thinking;
- Increased risk of academic dishonesty and reduced transparency;
- Ethical issues related to data privacy, authorship, and algorithmic bias.

Traditional evaluation methods, which emphasize manual grading and static rubrics, are ill-suited for this new landscape. As AI becomes an embedded component of student work, new evaluation paradigms must be developed to assess innovation, creativity, and problem-solving in authentic, AI-influenced learning environments.

1.2 Research Objectives and Contributions

In light of these challenges, this study proposes a multi-dimensional intelligent evaluation framework for assessing student assignment innovation in the era of generative AI. The framework is designed to capture nuanced learning outcomes in settings where human and AI contributions are interwoven.

The main objectives of this research are:

To design an evaluation system that combines structured Prompt engineering, NLP tools, and expert review to assess student assignments across cognitive and technical dimensions;

To empirically test this system in a controlled experiment involving 120 undergraduate engineering students working with and without AI assistance;

To analyze the impact of AI integration on four key evaluation dimensions: creativity, problem-solving ability, technological application, and logical rigor;

To propose design implications for future educational evaluation systems that are robust, ethical, and adaptive to AI-supported learning environments.

This study aims to bridge the gap between educational evaluation theory and the practical demands of intelligent, data-rich classrooms. It contributes to the development of engineering-relevant, AI-compatible assessment models, supporting more accurate, fair, and future-oriented academic evaluation in higher education.[1]

2. RELATED WORK

2.1 Generative AI in Education: Applications and Concerns

Generative artificial intelligence (GAI) technologies such as ChatGPT and GitHub Copilot have rapidly gained traction in education, enabling students to automate writing, code generation, and content ideation tasks [2]. These tools offer significant benefits, including reduced task load, improved fluency of expression, and access to high-quality linguistic and conceptual scaffolding [3].

However, their widespread use also introduces new concerns around academic integrity, originality, and independent thinking [4][5]. Researchers warn that excessive dependence on generative tools may undermine the development of critical and creative competencies in learners [6].Scholars have emphasized the importance of balancing generative tool use with deliberate design thinking strategies to maintain cognitive depth and originality in student work [7].

Educational institutions are thus forced to reevaluate not only content delivery, but also assignment formats and assessment criteria [8].

2.2 Human-AI Collaboration and Assignment Innovation

In recent years, the paradigm of human–AI collaboration has emerged in education, where students actively co-create with AI systems to complete open-ended tasks such as essays, design projects, or programming assignments [9]. Studies show that AI can provide real-time feedback, stimulate divergent thinking, and support individualized exploration [10].

Nevertheless, such collaboration demands new frameworks for measuring students' authentic contributions, as traditional rubrics often fail to capture co-generated outcomes or shared cognitive processes [11]. Moreover, qualitative studies have emphasized the need to differentiate between "AI-assisted originality"and "automated replication"in assignment outputs [12]. Recent frameworks advocate for the inclusion of innovation transparency indicators to distinguish authentic co-creation from mere algorithmic reproduction [13].

2.3 Intelligent Evaluation Systems and Multi-dimensional Indicators

To respond to these changes, researchers have proposed intelligent assessment systems that incorporate natural language processing (NLP), structured Prompt engineering, and machine learning to automate feedback and scoring [14]. These systems can evaluate grammar, logic, topical relevance, and even emotional tone [15]. More advanced models integrate personalization features and feedback loops, adjusting evaluation based on learning history and progress [16].

In parallel, there is a growing consensus on adopting multi-dimensional evaluation frameworks, which assess not only content accuracy, but also creativity, problem-solving ability, technological application, and cognitive rigor [17][18]. Tools such as the Alternative Uses Task (AUT) and problem-based project assessments have been widely used to measure creativity in education.

2.4 Research Gap and Study Positioning

Despite promising developments, two key research gaps remain. First, few studies have evaluated the impact of AI-assisted assignments in controlled experimental settings, especially in STEM-related practical tasks. Second, most existing evaluation systems remain tool-centered, lacking theoretical grounding in collaborative learning or innovation assessment.

This study addresses these gaps by proposing a structured, AI-compatible, multi-dimensional evaluation system for student assignments. The system is empirically tested in an engineering education context using both quantitative and qualitative methods, contributing to the field by offering a validated framework for evaluating innovation under human–AI collaboration.

3. SYSTEM FRAMEWORK

The design of an AI-based multi-dimensional evaluation system for student assignments must address both technological integration and pedagogical validity. To support this, we present a dual-layered system framework: one grounded in a practical ecosystem of generative AI tools (Table 1), and the other describing the logical architecture of the evaluation engine (Figure 2).

3.1 Ecosystem of Generative AI Tools in Education

Table 1. Generative AI Tools Categorized by Application Area

Application Area	AI Tool	Developer	Description
Education	iFlytek Spark, MathGPT	iFLYTEK, Tomorrow Advancing Life	Learning assistant platforms with subject-specific problem-solving capabilities
Image generation	MidJourney, DALL, Imagen, Stable Diffusion	Various	Tools that convert textual descriptions into visual content
Audio applications	Suno, Udio, HaiMan Music	Suno, ByteDance	AI tools for music generation, including Chinese music composition
Code generation	GitHub Copilot, CodeGeeX, Codium, IntelliCode	OpenAI, Tsinghua, Codium	Code completion, debugging, and generation assistants

Generative AI tools vary by application domain and functionality. In educational settings, tools such as iFlytek Spark and MathGPT provide content-specific support for learning and problem-solving, while others like GitHub Copilot assist

in code generation. These tools represent distinct categories such as text generation, image generation, audio synthesis, and structured feedback.

Table 1 summarizes key generative AI tools, their developers, and primary application areas, which form the external ecosystem influencing assignment evaluation tasks.

These tools highlight the diversity of inputs and modalities that modern students may engage with when producing assignments, and therefore must be accommodated within a flexible evaluation system.

3.2 Logical Architecture of the Evaluation System

The internal framework of the system is structured around four core modules: assignment input, AI content analysis, evaluation engine, and feedback generation. The process flow is illustrated in Figure 2.

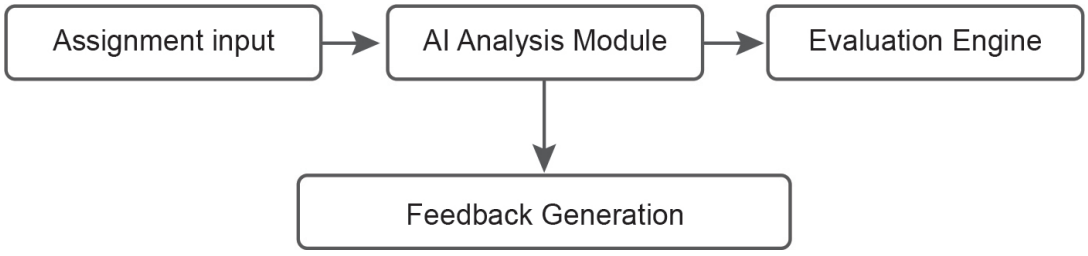


Figure 2. Simplified Logical Framework of AI-Based Assignment Evaluation

The evaluation system follows a linear yet modular structure, enabling real-time input analysis, structured scoring, and adaptive feedback delivery.

3.2.1 Assignment Input

Assignments are submitted via structured interfaces using predefined formats such as Markdown or YAML, allowing precise parsing and metadata extraction.

3.2.2 AI Analysis Module

This module performs topic extraction, semantic matching, originality detection, and creativity estimation using NLP and AI classification models.

3.2.3 Evaluation Engine

Based on a multi-dimensional rubric, the engine computes scores for innovation, technical use, problem-solving, and logical structure.

3.2.4 Feedback Generation

Automated yet contextual feedback is generated in human-readable form, supporting student reflection and revision. The system stores anonymized learning data for model optimization.

Together, the ecosystem and internal logic form a comprehensive infrastructure for intelligent evaluation of student assignments in AI-augmented learning environments.

4. METHODOLOGY

To explore how generative artificial intelligence can be effectively applied to evaluate student assignment innovation, this study adopts a mixed-methods research design. The approach combines structured input formatting, AI-driven content analysis, and multidimensional scoring supported by both machine intelligence and expert judgment. The methodology consists of three key components: experimental design, structured prompt workflow, and system implementation logic.

4.1 Experimental Design and Participants

The experiment was conducted at Shanghai University, involving 120 second-year undergraduate students majoring in computer engineering. Participants were randomly assigned into two groups: the experimental group (n = 60), which used ChatGPT for assistance, and the control group (n = 60), which completed assignments independently.

The assignment topic for both groups was identical: "Design a Smart Home Control System." Deliverables included system architecture, functional design, code implementation, and technical documentation. To control for content variation, all submissions were collected via a unified digital platform with a standardized format.

Each assignment was evaluated across four core dimensions: creativity and innovation, problem-solving ability, technological application, and logical rigor. The scoring process involved AI-based automated evaluation and instructor-led validation to ensure both efficiency and credibility.

4.2 Prompt-Based Workflow and Logical Architecture

To standardize input and enhance evaluability, students in the experimental group submitted their assignments using structured prompts, formatted in YAML or JSON. These prompts captured not only the assignment content but also relevant metadata, such as self-identified innovation type, programming languages used, and design goals.

The system's operational workflow is illustrated in Figure 3, showing the flow from input collection to feedback generation.

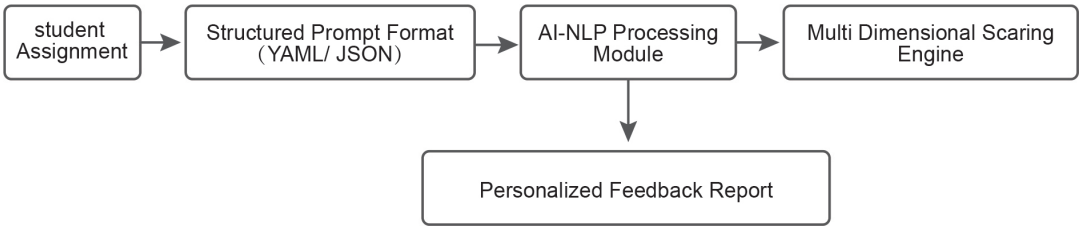


Figure 3. Workflow of Structured Prompt-Driven Evaluation Process

Assignments are submitted through structured prompts, parsed by the AI-NLP engine, scored across multiple dimensions, and returned with personalized feedback reports.

The workflow begins with the submission of student work encoded in structured prompts. These are first parsed and pre-processed using an NLP engine that extracts relevant linguistic and technical features. Key indicators such as originality, fluency, keyword density, and topical diversity are computed.

Next, the multi-dimensional scoring engine maps these features onto the four evaluation dimensions using pre-defined scoring models. Finally, the system generates a personalized feedback report, highlighting strengths, weaknesses, and actionable improvement suggestions for each dimension.

This architecture ensures that the evaluation process is standardized, scalable, and adaptive, making it well-suited for assignments generated or co-authored with AI tools.

4.3 Data Collection and Performance Indicators

To assess the effectiveness of the system, both quantitative and qualitative data were collected. The primary quantitative indicators included average scores per evaluation dimension, group-based comparison statistics (e.g., t-tests, p-values), and keyword frequency distributions derived from assignment text mining.

Complementing this, qualitative data were gathered through open-ended student reflections, instructor annotations, and system interaction logs. These data were analyzed to capture user perceptions, ethical considerations, and engagement patterns when working with generative AI tools. Ethnographic and interactional data offer critical insight into how learners interpret and appropriate generative outputs, reflecting the evolving boundaries of authorship and agency in AI-supported education [19].

All statistical analyses were conducted using SPSS 27.0, while qualitative content coding was supported by NVivo 12.

Through this methodology, the study not only investigates the impact of generative AI on assignment innovation, but also proposes a replicable, technically grounded evaluation framework for AI-augmented learning environments.

5. EXPERIMENT AND RESULTS

To evaluate the practical effectiveness of the proposed AI-assisted evaluation framework, a comparative experiment was conducted involving two student groups: one using generative AI tools during assignment completion, and a control group working without AI assistance. The analysis focused on four key evaluation dimensions: innovation, problem-solving ability, technology use, and logical rigor.

5.1 Quantitative Results

The evaluation scores of both groups were analyzed using SPSS 27.0. As shown in Figure 4, the experimental group consistently outperformed the control group across all core dimensions.

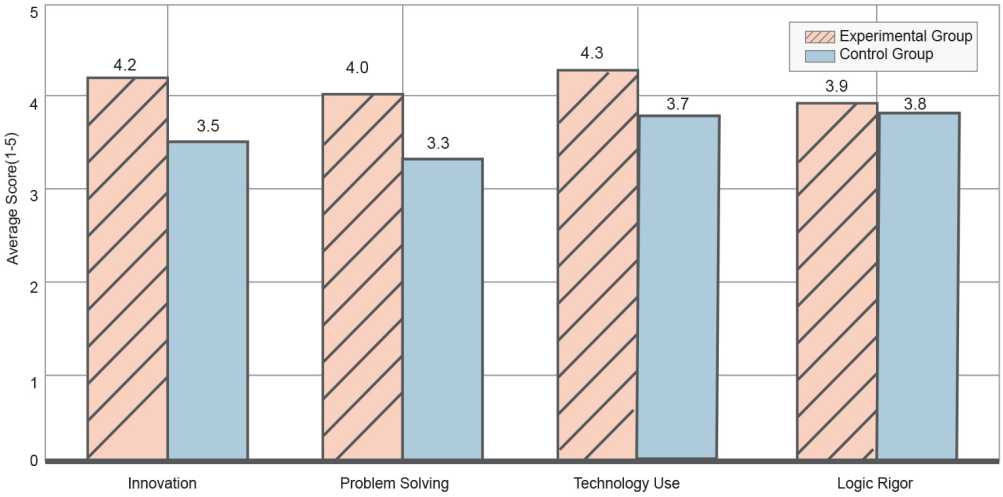


Figure 4. Comparison of Evaluation Dimensions Between Groups

The bar chart illustrates the average scores (scale: 1–5) for each evaluation dimension in both the experimental and control groups.

Specifically:

Innovation: The experimental group achieved a mean score of 4.2, compared to 3.5 in the control group, showing a significant improvement attributed to AI-assisted ideation.

Problem Solving: Students in the AI-assisted group scored 4.0, while the control group scored 3.3, indicating enhanced strategic thinking when supported by generative tools.

Technology Use: A notable advantage was seen here, with the experimental group reaching 4.3 versus 3.7 in the control group, suggesting greater technical sophistication in implementation.

Logical Rigor: The difference was marginal (3.9 vs. 3.8), indicating that while AI support enhances creativity and implementation, it may not substantially affect formal logic structuring.

These results confirm that AI-supported learning environments are particularly effective in boosting creative output, applied technology use, and problem-solving capacity, though the impact on logical composition remains less pronounced.

5.2 Statistical Significance and Interpretation

Independent-samples t-tests revealed the following:

- Innovation ($p < 0.01$): Statistically significant;
- Problem Solving ($p < 0.05$): Statistically significant;
- Technology Use ($p < 0.01$): Statistically significant;

- Logic Rigor ($p > 0.05$): Not statistically significant.

This validates that the proposed evaluation system can capture meaningful differences in assignment quality when generative AI is used as a design collaborator. These differences are most prominent in non-linear thinking and technical adaptability, aligning with the theoretical expectation that AI extends ideation and implementation capabilities.

5.3 Additional Observations

Text mining analysis of assignment reports revealed that students in the experimental group used AI-related terms (e.g., "machine learning,""API integration,""ChatGPT optimization") 2.5×more frequently than those in the control group. Interaction logs further indicate that students predominantly used ChatGPT during the early ideation phase (45%) and technical coding implementation phase (38%), suggesting that AI played a dual role as a creative assistant and coding collaborator.

Combined, these quantitative and qualitative results demonstrate that integrating generative AI into assignment workflows not only enhances output quality but also influences the way students approach problem-solving and expression.

6. CONCLUSION AND FUTURE WORK

This study proposes and validates a multi-dimensional evaluation framework for student assignment innovation in the context of generative artificial intelligence (GAI). By integrating structured prompts, AI-assisted analysis, and human-in-the-loop scoring, the framework enables efficient, scalable, and pedagogically sound assessment across creativity, problem-solving, technology use, and logical rigor.

Through a controlled experiment involving 120 computer engineering undergraduates, the results demonstrate that GAI-supported students significantly outperform their peers in innovation ($p < 0.01$), technical application ($p < 0.01$), and problem-solving ability ($p < 0.05$). These findings confirm the positive role of GAI as a creative collaborator and technical enabler in student-centered learning environments. At the same time, the minimal improvement in logic rigor suggests that critical reasoning and structural coherence remain largely dependent on human cognitive input.

The study also highlights how structured prompt formats and AI-NLP processing can support transparent, reproducible, and personalized evaluation workflows. This system balances automation and interpretability, helping educators adapt to the challenges of academic authenticity, cognitive equity, and feedback scalability in the age of intelligent education.

Despite its contributions, the research has several limitations. The system was tested within a single institution and discipline, and the AI model used was task-specific. Further generalization requires broader samples, subject integration, and cross-linguistic adaptability. In addition, issues of data privacy, AI-generated content attribution, and over-reliance on automation merit continuous monitoring.

Future work will focus on three directions:

- refining real-time adaptive scoring algorithms through deep learning and reinforcement feedback;
- incorporating advanced technologies such as blockchain and federated learning to ensure evaluation traceability and data security;
- extending the framework to support multilingual and cross-cultural assignment evaluation, particularly for globalized educational scenarios and collaborative innovation tasks.

In conclusion, this study provides a replicable technical and pedagogical foundation for intelligent evaluation systems in AI-augmented learning. By aligning engineering tools with human-centered education values, it supports the sustainable transformation of assessment in the era of generative intelligence.

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